

Section 93 — Master Computational Number Theory

Mixed Contest

Contest contract

Six original problems combine the tools of Sections 91–92.

Problem	Main structure
A	certified leaves of Pollard–rho recursion
B	prime-power Carmichael values joined by LCM
C	factor and reduce a candidate group exponent
D	primitive-root coordinates plus a linear congruence
E	repeated Tonelli–Shanks queries
F	discrete log followed by an affine congruence

The important habit is to separate layers. First discover or certify the group order. Then translate exponentiation into a congruence. Only after that solve the remaining linear arithmetic.

Layered derivation

Suppose a primitive root g is available modulo prime p . Writing every nonzero value as a power of g changes multiplication into addition and exponentiation into multiplication of exponents modulo $p-1$. An equation such as $x^k=a$ becomes

$$k * \log_g(x) = \log_g(a) \text{ modulo } p-1.$$

This is now a linear congruence. It has solutions exactly when the gcd of k and $p-1$ divides the right side, and all exponent solutions must then be translated back with powers of g . Keeping the discrete-log and congruence layers separate makes both the proof and no-solution cases visible.

Carmichael’s function and multiplicative order have a similar hierarchy. $\lambda(n)$ is an exponent known to work for every unit; the order of one unit divides it. Factor the candidate exponent, tentatively remove one prime factor at a time, and keep the removal precisely when modular exponentiation still returns one.

Contest audit

- Are all factorization leaves certified prime?
- Is the value a unit before group arguments or inverses are used?
- Is the relevant modulus p , $p-1$, $\phi(n)$, or $\lambda(n)$?
- Does the task ask for any solution, the smallest exponent, or all roots?
- Can multiplication, lcm, or an exponent overflow before modular reduction?

Brute force complete residue systems for small moduli. It simultaneously tests existence, minimality, multiplicity, and conversion back from exponent space.

Exercises

A–F form the complete mixed contest.